VIKTOR CHALLENGE · TUM.AI · BUILD THE ROUTER

routemaxxing

−45.8% estimated cost on 313 held-back trajectories, proxy failure -0.19 pp

Atahan Kılıç · Musa Kocabasoğlu · Utku Kalkanlı

OBJECTIVE

Cheapest route that clears an honest quality bar

Minimize cache-aware estimated dollars while the one-sided 95% upper bound on proxy-failure degradation stays within **2 percentage points**. We report a frontier, not a point.

THE SIGNAL

Tool dialect fixes the family; call zero picks the model

Structure in the traces

Claude rows expose `bash`, `file_read`, `file_edit`; GPT rows expose `shell_command`, `apply_patch`. Across 1,000 rows the two sets **never overlap**, so a cross-family swap is an action the log never took. We route only within a family, on call-zero features: trigger, intent, prompt size, image, toolset.

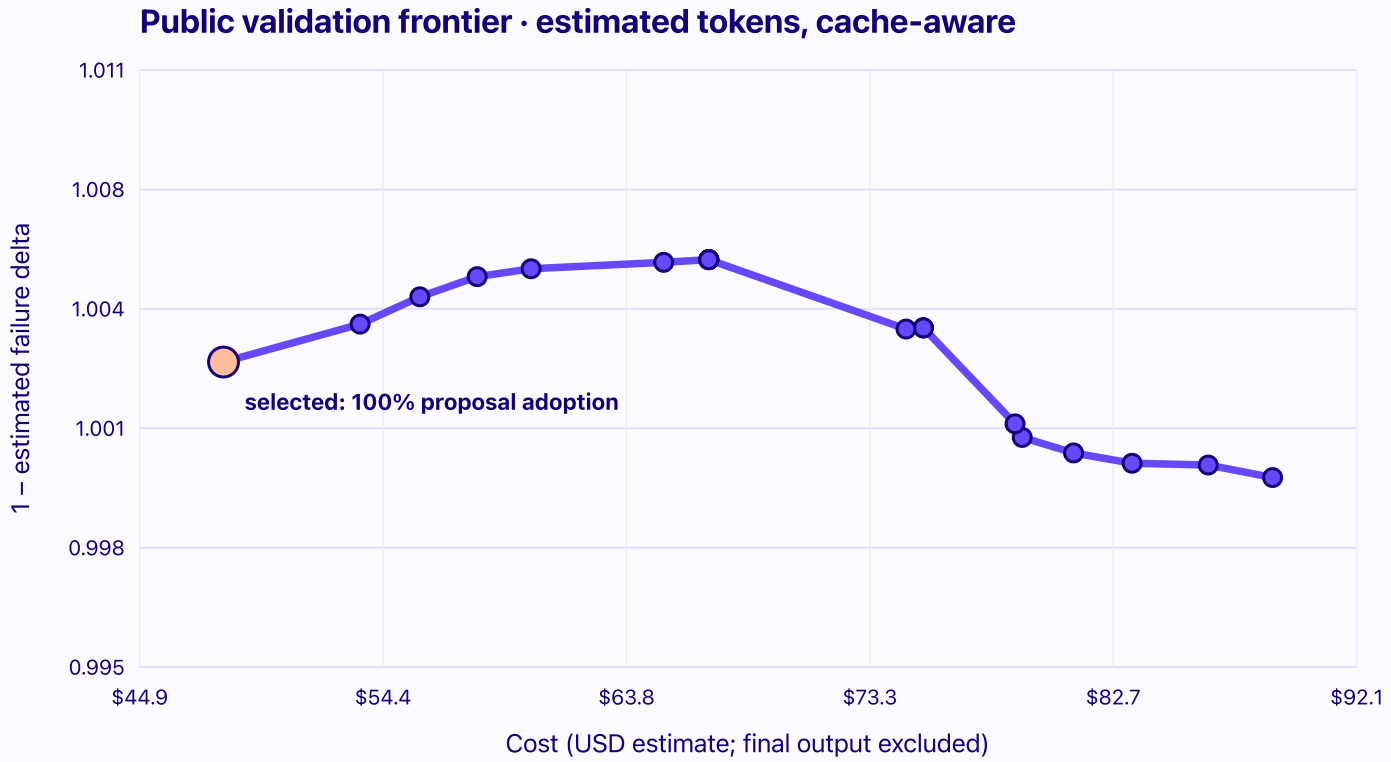
Shown on a real trace

Trajectory `2e9ea81ffd3a` — 11 calls on the Claude dialect. Logged **claude-opus-5** at $0.51; Routemaxxing picks **claude-sonnet-5** at $0.20 and holds it for the whole trajectory, so the prefix cache is never reset.

The whole policy is **three moves**: opus-5→sonnet-5 **331** · sol→terra **112** · fable-5→sonnet-5 **71** — 514 of 1,000 trajectories routed, the rest left alone.

THE ONE CHART

Cost–proxy-quality frontier (public validation; organizer held-out pending)



HONESTY

What the log can — and cannot — identify

Method — Family-restricted propensity screen + outcome regression on downgrade-from-default; 2,000 trajectory bootstraps. Policy locked on train, reported on 313 untouched trajectories.

Cache trap — Cached share is inferred from item-level prefix overlap; a model switch resets it and the next call repays the full prefix. Our certified route switches **zero** times mid-trajectory, so it never pays that penalty.

Support — Routed ESS 155.4, minimum target propensity 0.255. Luna has 12 training rows against a floor of 20, so the router never selects it.

Known weakness — Quality is an **automatic proxy** (unresolved tool error, repeated action), not a measured label — and a cheaper early model could have produced a different later context, which our estimate cannot observe.

CLOSE

Half the spend on this log bought nothing the proxy can detect.

$88.88 → $48.18 on held-back trajectories · 58% routed

Weakness: the quality bar is a proxy, not a label. Next: run once on the organizer held-out split — and do not retune.